U.S. Antimony Supply Resilience

Strategic options for a hypothetical defense manufacturer, 2027–2031

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Research cutoff: 4 November 2026

Illustrative procurement model · 8-slide edition

DECISION RECOMMENDATION

Qualify a second supply lane, then hold five months of stock

Conditional balanced policy for 100 t/year of contained antimony-equivalent demand

45%

Target share from an independently qualified alternative international supply lane

5 months

Provisional buyer-qualified reserve: $2.30m at the historical USGS metal proxy

30%

Future domestic offtake target; availability begins only with verified qualified output

2030 severe interruption

0 t unfilled under modeled supplier assumptions

2027–29: the modeled minimum buffer is 4.91 months for ≤5 t unfilled. Five months still leaves 4.35 t unfilled before domestic qualification.

USGS historical price proxy; model inputs and assumptions are documented in the public evidence register.

Evidence and assumptions

SUPPLY BASELINE

National import reliance does not establish a buyer's supplier mix

USGS estimates and historical U.S. antimony import origin, different denominators

91%

U.S. net import reliance, 2025 estimate

55%

China share of U.S. metal + oxide import origins, 2021–2024

BUYER-SPECIFIC DILIGENCE

Form + grade	Contained-Sb conversion and specification
Traceability	Mine, smelter and refinery location
Qualification	Test results, supply independence and acceptance

China's item-specific export licensing is a separate sourcing gate. EXIM's $2.906bn Stibnite authorization and an INL pilot still require buyer-level qualification evidence.

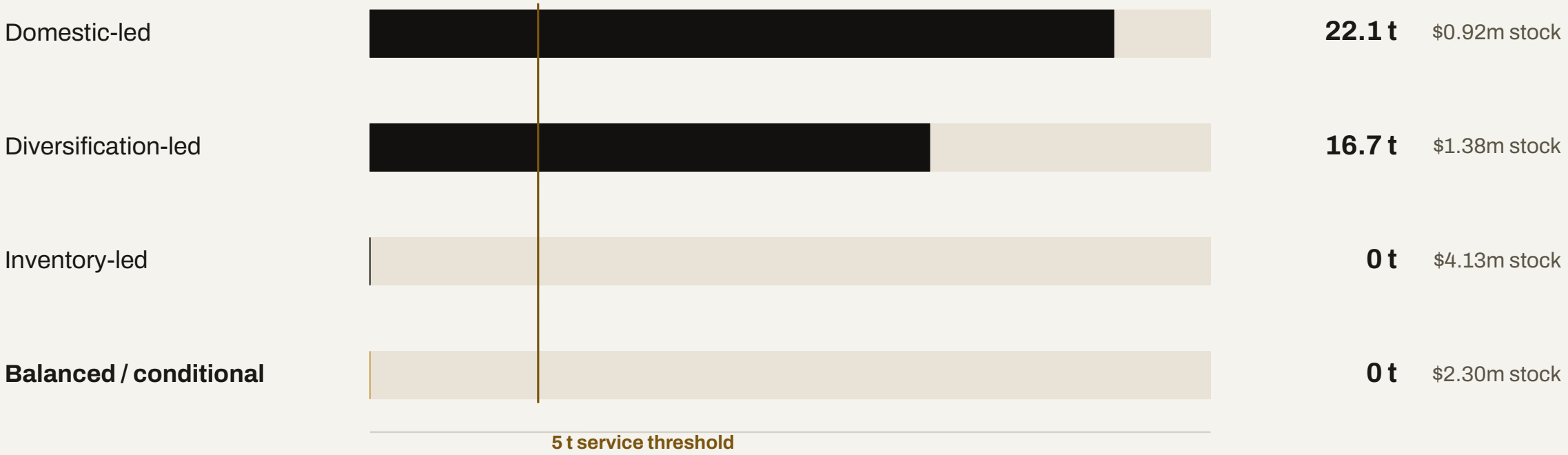
USGS MCS 2026 · EXIM transaction register · INL pilot · MOFCOM No. 33

FOUR-STRATEGY COMPARISON

Only the balanced strategy meets both modeled 2030 constraints

Severe eight-month disruption; 100 t/year buyer; cash cap $3m; service tolerance ≤5 t unfilled

UNFILLED ANTIMONY-EQUIVALENT (T)



Inventory-led passes continuity but exceeds the illustrative $3m stock limit; balanced passes both only under specified qualification assumptions.

Model register / 2030 comparison

ECONOMIC TRADE-OFF

Five months frees $1.84m of stock cash versus nine months

2030 severe case; annual incremental expense includes sourcing premiums, carry and spot markup

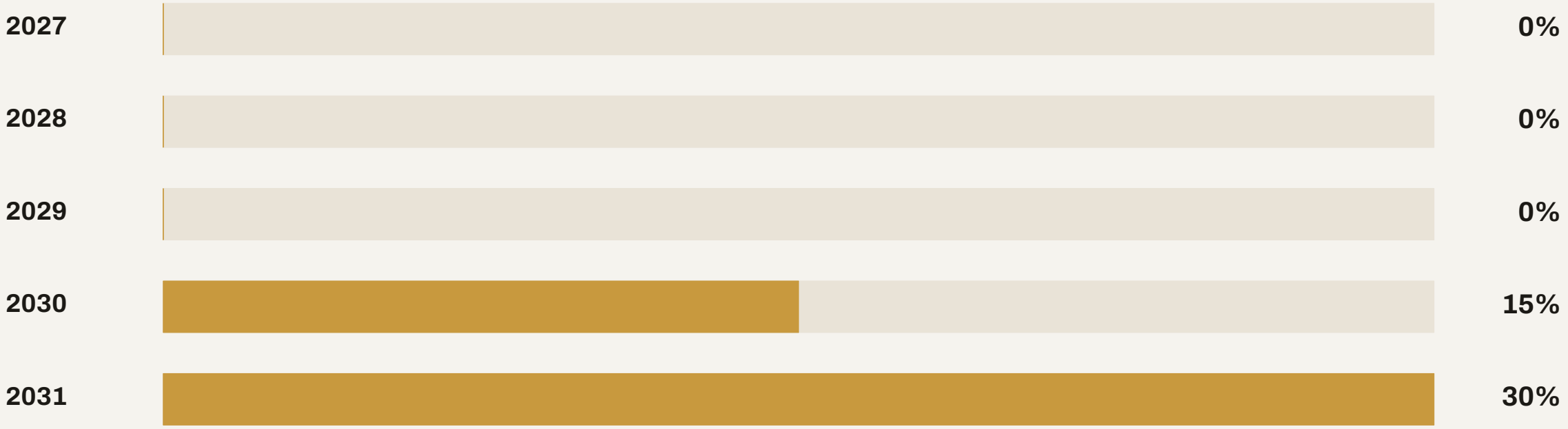
	BALANCED / CONDITIONAL	INVENTORY-LED
Stock months	5 months	9 months
Inventory capital	$2.296m	$4.134m
Modeled annual increment	$813k	$703k
2030 residual unfilled	0 t	0 t

Trade-off: lower initial working capital at approximately $110k more annual modeled expense

Initial inventory cash is an asset, not an annual incremental expense; all amounts use hypothetical buyer and historical price inputs.

The domestic target becomes available only after qualification

Modeled ramp to 30% target share; all values are illustrative buyer sourcing shares



Until qualified domestic deliveries are available, that target volume is assigned to the incumbent sourcing lane in the model.

Actual schedules, plant status, processing specifications and enforceable offtake terms would control a real procurement ramp.

REVERSAL CONDITIONS

The five-month policy is sensitive to supplier independence

Residual unfilled contained-Sb equivalent in each distinct adverse scenario

SEVERE TEST	MODELED GAP	OUTCOME AGAINST ≤5 t
2027: no qualified alternative	16.50 t	Above tolerance
2030: alternative lane 75% curtailed	5.70 t	Above tolerance
2030: curtailment + domestic delay	12.45 t	Above tolerance
2030: 12-month interruption	15.15 t	Above tolerance
Action: strengthen alternate refinery qualification and retain a sixth-month procurement option		

Each test is a separate modeled shock, not an expected forecast or cumulative five-year loss.

EXECUTION GATES

Procurement, Finance and Legal each control a material assumption

Decision rights before firming purchases, inventory funding or domestic offtake

OWNER	VERIFICATION REQUIRED	TRIGGER
Procurement + Engineering	Grade, chain of custody, independent refining, delivery tests	Qualify alternative 45% lane
Finance	Actual quoted price, carry, storage, working-capital ceiling	Approve five-month reserve
Strategy + Legal	Financing close, processing stage, product acceptance, contract remedies	Stage conditional domestic offtake

USGS MCS 2026 · EXIM register · Perpetua 10-Q · INL pilot · MOFCOM No. 33

MOFCOM No. 46 · MOFCOM No. 72 · Model register · USAC 10-Q · DLA stockpile

ILLUSTRATIVE BUYER CASE · SOURCE CUTOFF 4 NOV 2026 · INDEPENDENT SMPT ANALYSIS

Read case / model